

Derivation of the Full Width at Half Maximum (FWHM) for a Gaussian Distribution

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AST 4762C Homework 4 Question 3.a

Question

Show analytically that the full width at half maximum (FWHM) of a Gaussian distribution is 2.354σ (see Bevington, p. 28).

Derivation

The general equation for a one-dimensional Gaussian probability density function with mean μ and standard deviation (width) σ is given by:

$$p(x) = \frac{e^{(-\frac{(x-\mu)^2}{2\sigma^2})}}{\sigma\sqrt{2\pi}}$$

The maximum value of the distribution occurs at the center of the peak where $x = \mu$:

$$p_{max} = p(\mu) = \frac{e^0}{\sigma\sqrt{2\pi}} = \frac{1}{\sigma\sqrt{2\pi}}$$

To find the two positions x where the function drops to half of its maximum value, we set $p(x) = \frac{1}{2}p_{max}$:

$$\frac{e^{(-\frac{(x-\mu)^2}{2\sigma^2})}}{\sigma\sqrt{2\pi}} = \frac{1}{2} \left(\frac{1}{\sigma\sqrt{2\pi}} \right)$$

Cancel the normalization factor $\frac{1}{\sigma\sqrt{2\pi}}$ from both sides:

$$e^{(-\frac{(x-\mu)^2}{2\sigma^2})} = \frac{1}{2}$$

Apply the natural logarithm ($\ln$) to both sides to eliminate the exponential function:

$$-\frac{(x - \mu)^2}{2\sigma^2} = \ln\left(\frac{1}{2}\right) = \ln(2^{-1}) = -\ln(2)$$

Multiply both sides by $-2\sigma^2$:

$$(x - \mu)^2 = 2\sigma^2 \ln(2)$$

Take the square root of both sides, yielding two points (x_1 on the left and x_2 on the right of the mean):

$$x - \mu = \pm \sigma \sqrt{2 \ln(2)}$$

$$x_{1,2} = \mu \pm \sigma \sqrt{2 \ln(2)}$$

The FWHM is the distance between these two points ($x_2 - x_1$):

$$\text{FWHM} = x_2 - x_1 = (\mu + \sigma \sqrt{2 \ln(2)}) - (\mu - \sigma \sqrt{2 \ln(2)})$$

$$\text{FWHM} = 2\sigma \sqrt{2 \ln(2)}$$

Substituting the value of $\ln(2) \approx 0.693147$:

$$\text{FWHM} = 2\sigma \sqrt{2 \times 0.693147} = 2\sigma \sqrt{1.38629}$$

$$\text{FWHM} \approx 2.354\sigma$$